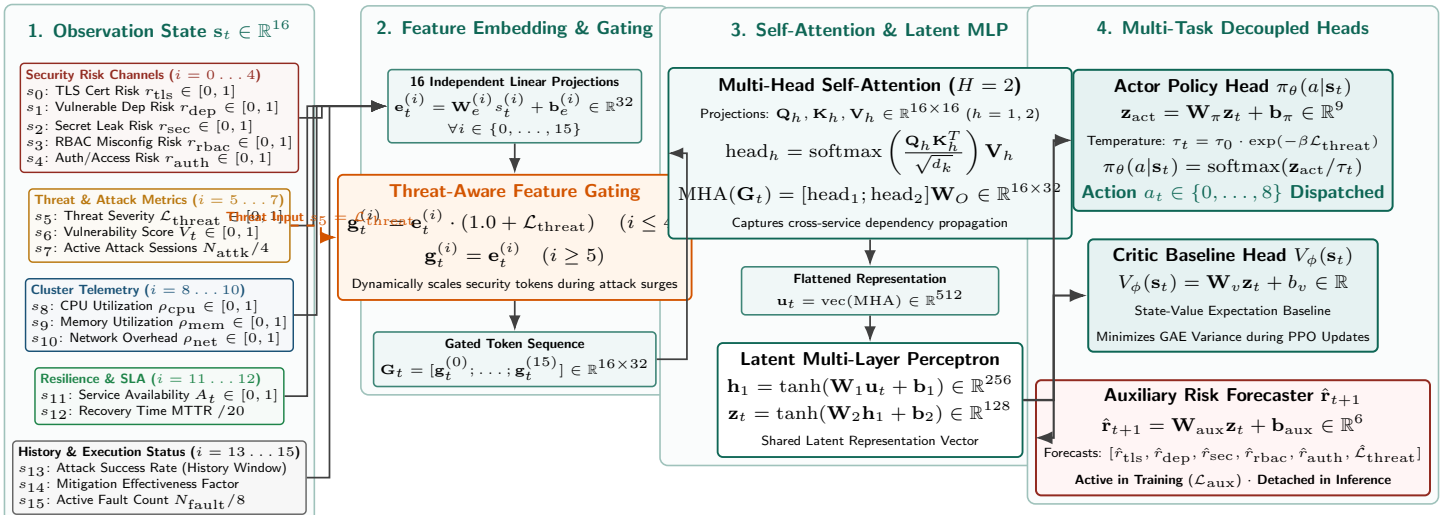

1. Observation State $\mathbf{s}_t \in \mathbb{R}^{16}$
Security Risk Channels ($i = 0 \ldots 4$)
s_0: TLS Cert Risk $r_{\text{tls}} \in [0, 1]$
s_1: Vulnerable Dep Risk $r_{\text{dep}} \in [0, 1]$
s_2: Secret Leak Risk $r_{\text{sec}} \in [0, 1]$
s_3: RBAC Misconfig Risk $r_{\text{rbac}} \in [0, 1]$
s_4: Auth/Access Risk $r_{\text{auth}} \in [0, 1]$
Threat & Attack Metrics ($i = 5 \ldots 7$)
s_5: Threat Severity $\mathcal{L}_{\text{threat}} \in [0, 1]$
s_6: Vulnerability Score $V_t \in [0, 1]$
s_7: Active Attack Sessions $N_{\text{attk}}/4$
Cluster Telemetry ($i = 8 \ldots 10$)
s_8: CPU Utilization $\rho_{\text{cpu}} \in [0, 1]$
s_9: Memory Utilization $\rho_{\text{mem}} \in [0, 1]$
s_{10}: Network Overhead $\rho_{\text{net}} \in [0, 1]$
Resilience & SLA ($i = 11 \ldots 12$)
s_{11}: Service Availability $A_t \in [0, 1]$
s_{12}: Recovery Time MTTR $/20$
History & Execution Status ($i = 13 \ldots 15$)
s_{13}: Attack Success Rate (History Window)
s_{14}: Mitigation Effectiveness Factor
s_{15}: Active Fault Count $N_{\text{fault}}/8$
2. Feature Embedding & Gating
16 Independent Linear Projections
$\mathbf{e}_t^{(i)} = \mathbf{W}_e^{(i)} s_t^{(i)} + \mathbf{b}_e^{(i)} \in \mathbb{R}^{32}$
$\forall i \in \{0, \ldots, 15\}$
Threat-Aware Feature Gating
Threat Input $s_5 = \mathcal{L}_{\text{threat}} \in [0, 1]$
$\mathbf{g}_t^{(i)} = \mathbf{e}_t^{(i)} \cdot (1.0 + \mathcal{L}_{\text{threat}}) \quad (i \leq 4)$
$\mathbf{g}_t^{(i)} = \mathbf{e}_t^{(i)} \quad (i \geq 5)$
Dynamically scales security tokens during attack surges
Gated Token Sequence
$\mathbf{G}_t = [\mathbf{g}_t^{(0)}; \ldots; \mathbf{g}_t^{(15)}] \in \mathbb{R}^{16 \times 32}$
3. Self-Attention & Latent MLP
Multi-Head Self-Attention ($H = 2$)
Projections: $\mathbf{Q}_h, \mathbf{K}_h, \mathbf{V}_h \in \mathbb{R}^{16 \times 16}$ $(h = 1, 2)$
$\text{head}_h = \text{softmax}\left(\frac{\mathbf{Q}_h \mathbf{K}_h^T}{\sqrt{d_k}}\right) \mathbf{V}_h$
$\text{MHA}(\mathbf{G}_t) = [\text{head}_1; \text{head}_2]\mathbf{W}_O \in \mathbb{R}^{16 \times 32}$
Captures cross-service dependency propagation
Flattened Representation
$\mathbf{u}_t = \text{vec}(\text{MHA}) \in \mathbb{R}^{512}$
Latent Multi-Layer Perceptron
$\mathbf{h}_1 = \tanh(\mathbf{W}_1 \mathbf{u}_t + \mathbf{b}_1) \in \mathbb{R}^{256}$
$\mathbf{z}_t = \tanh(\mathbf{W}_2 \mathbf{h}_1 + \mathbf{b}_2) \in \mathbb{R}^{128}$
Shared Latent Representation Vector
4. Multi-Task Decoupled Heads
Actor Policy Head $\pi_\theta(a|\mathbf{s}_t)$
$\mathbf{z}_{\text{act}} = \mathbf{W}_\pi \mathbf{z}_t + \mathbf{b}_\pi \in \mathbb{R}^9$
Temperature: $\tau_t = \tau_0 \cdot \exp(-\beta \mathcal{L}_{\text{threat}})$
$\pi_\theta(a|\mathbf{s}_t) = \text{softmax}(\mathbf{z}_{\text{act}}/\tau_t)$
Action $a_t \in \{0, \ldots, 8\}$ Dispatched
Critic Baseline Head $V_\phi(\mathbf{s}_t)$
$V_\phi(\mathbf{s}_t) = \mathbf{W}_v \mathbf{z}_t + b_v \in \mathbb{R}$
State-Value Expectation Baseline
Minimizes GAE Variance during PPO Updates
Auxiliary Risk Forecaster $\hat{\mathbf{r}}_{t+1}$
$\hat{\mathbf{r}}_{t+1} = \mathbf{W}_{\text{aux}} \mathbf{z}_t + \mathbf{b}_{\text{aux}} \in \mathbb{R}^6$
Forecasts: $[\hat{r}_{\text{tls}}, \hat{r}_{\text{dep}}, \hat{r}_{\text{sec}}, \hat{r}_{\text{rbac}}, \hat{r}_{\text{auth}}, \hat{\mathcal{L}}_{\text{threat}}]$
Active in Training ($\mathcal{L}_{\text{aux}}$) · Detached in Inference